

Manuscript figures and captions

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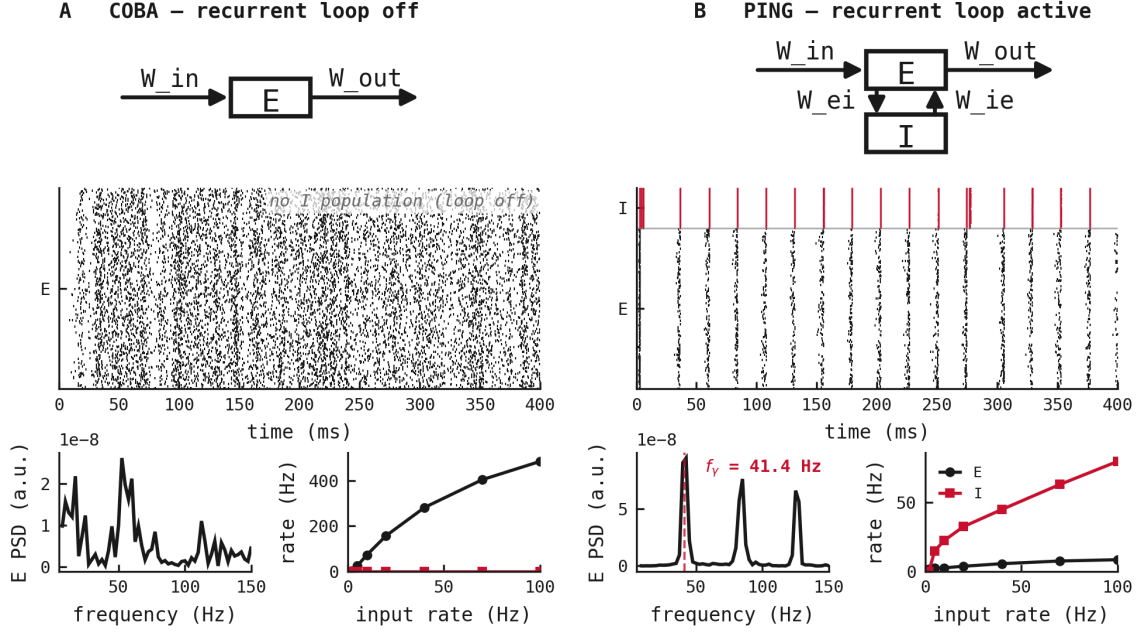


Figure 1: **A single recurrent $E \rightarrow I \rightarrow E$ loop simultaneously generates a gamma rhythm and clamps the excitatory firing rate.** Free-running activity of the two-population conductance-based network (excitatory pool $N_E = 1024$, inhibitory pool $N_I = 256$; canonical biophysical parameters, §5.1) under matched Poisson drive, in two configurations. Each column shows, from top: a wiring schematic, a single-trial spike raster, the excitatory power spectral density (PSD), and the excitatory $f-I$ curve. **(A)** COBA baseline with the recurrent loop disabled ($W^{EI} = W^{IE} = 0$): input projects to the excitatory (E) population only, with no inhibitory (I) population (schematic). The E spike raster is asynchronous, the Welch PSD of the summed E population shows no gamma-band peak, and the excitatory $f-I$ curve rises to ≈ 485 Hz under the strongest drive tested. **(B)** PING configuration with the loop engaged (schematic: $E \rightarrow I$ via W^{EI} , $I \rightarrow E$ via W^{IE} ; no $I \rightarrow I$ or $E \rightarrow E$ synapse): the inhibitory (I) population fires synchronous bursts, the E raster forms gamma bands, the PSD shows a discrete peak at $f_\gamma \approx 41$ Hz, and on axes shared with (A) the E rate is held approximately an order of magnitude lower across two decades of input drive while the I rate rises to ≈ 80 Hz. Source: exp023.

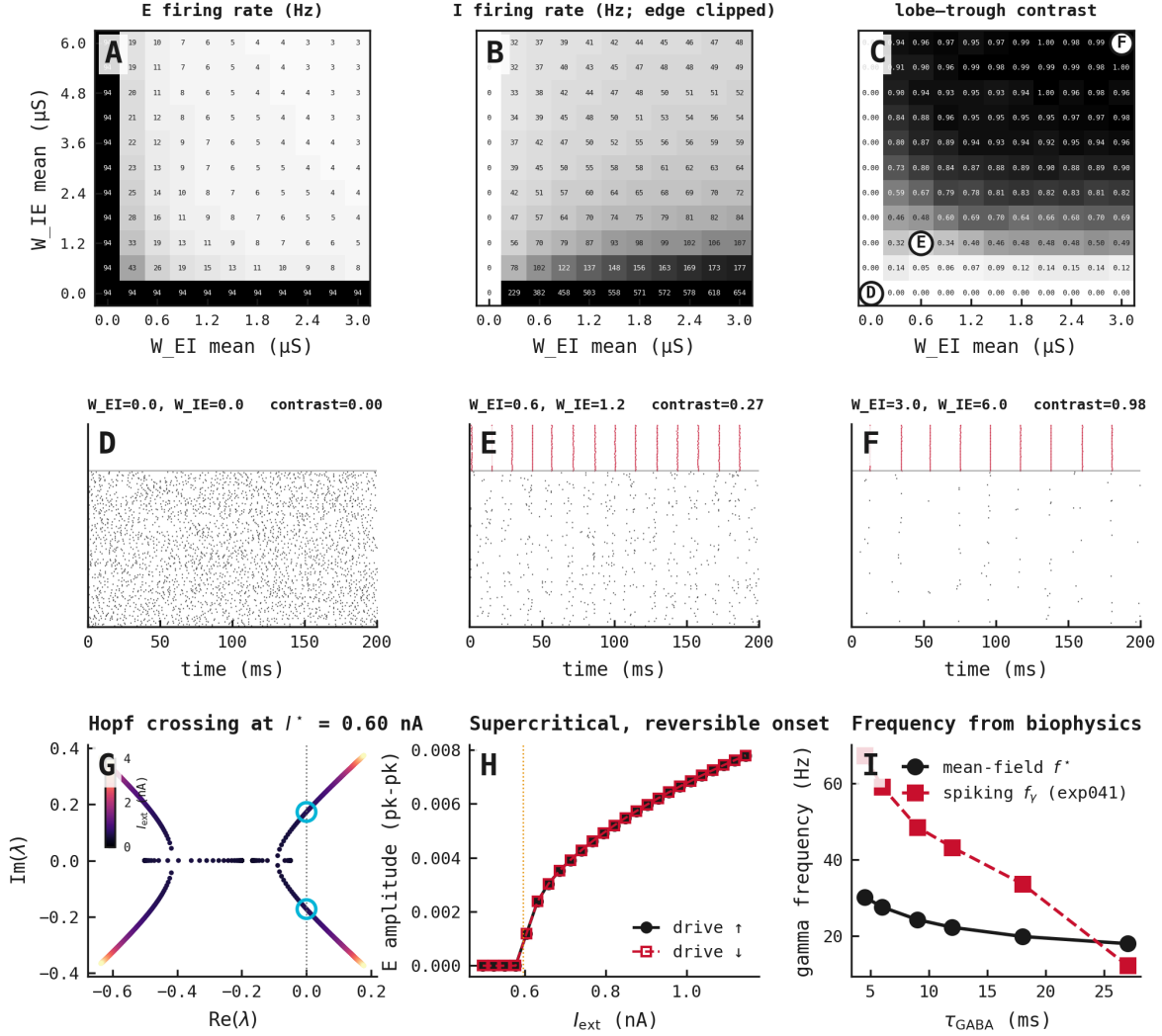


Figure 2: **Gamma emerges through a smooth, reversible onset across the coupling plane, consistent with a supercritical Hopf bifurcation of the mean-field reduction.** Nine panels (A–I). (A–C) Steady-state measurements across the 11×11 $W^{EI} \times W^{IE}$ coupling plane: mean excitatory firing rate (A), mean inhibitory firing rate (B), and the lobe–trough rhythmicity contrast R (C, §5.5), which is ≈ 0 along the two COBA edges and rises to ≈ 0.98 toward strong coupling. (D–F) Single-trial E (black) and I (red) rasters at three points sampled along the $W^{IE} = 2W^{EI}$ diagonal (circled in C): the loop-disabled origin (D, asynchronous), weak coupling ($R < 0.5$, E), and strong coupling (F, sharp gamma volleys). (G–I) The four-dimensional conductance mean-field reduction (§5.3): a complex-conjugate eigenvalue pair crosses into the right half-plane at external drive $I^* = 0.6$ nA (G), locating a Hopf bifurcation at $f^* \approx 27.6$ Hz; the steady-state oscillation amplitude grows continuously across the onset with coincident up- and down-ramp branches (H), the signature of a supercritical, reversible transition; and the predicted gamma frequency falls with τ_{GABA} in qualitative agreement with the spiking measurement (I). Source: exp054 (coupling-plane maps and mean-field, incorporating exp033); spiking f_γ from exp041.

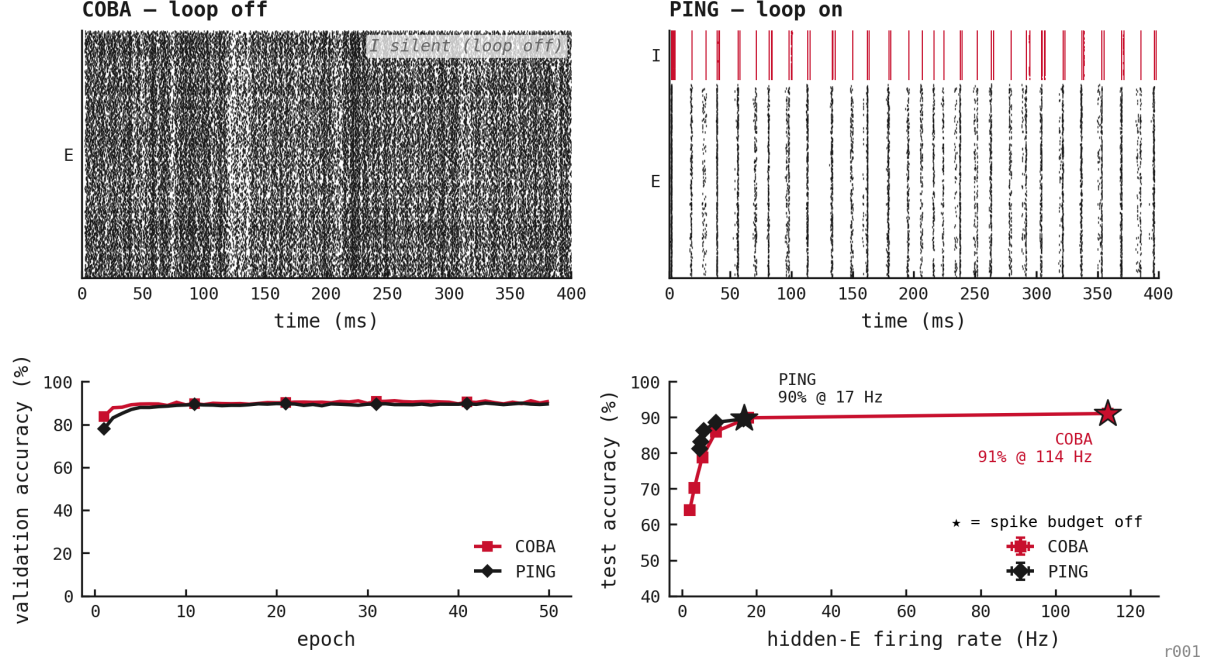


Figure 3: **Trained PING matches COBA classification accuracy while operating at an order-of-magnitude lower excitatory firing rate.** Both configurations were trained on MNIST by surrogate-gradient descent (§5.4). Top: representative single-trial rasters of the trained networks: COBA fires densely and asynchronously with the inhibitory population silent, whereas PING fires in gamma bands with synchronous inhibitory bursts (red) above excitatory spikes (black). For visualisation, each raster is an extended 400 ms replay of one digit, twice the 200 ms presentation used for training and quantitative evaluation. Bottom left: test accuracy per epoch, both configurations converging to $\approx 90\%$. Bottom right: accuracy–rate frontier traced by sweeping the sample-wise hidden-E rate ceiling (§5.4); each marker shows mean hidden-E firing rate (abscissa) and test accuracy (ordinate) across three independent seeds, with standard-error bars. PING lies above and to the left of COBA across the sweep. At the unpenalised operating point (starred), PING reaches $\approx 90\%$ at ≈ 16.6 Hz, against COBA’s $\approx 91\%$ at ≈ 114 Hz. Source: exp025 rate-attractor analysis in exp024.

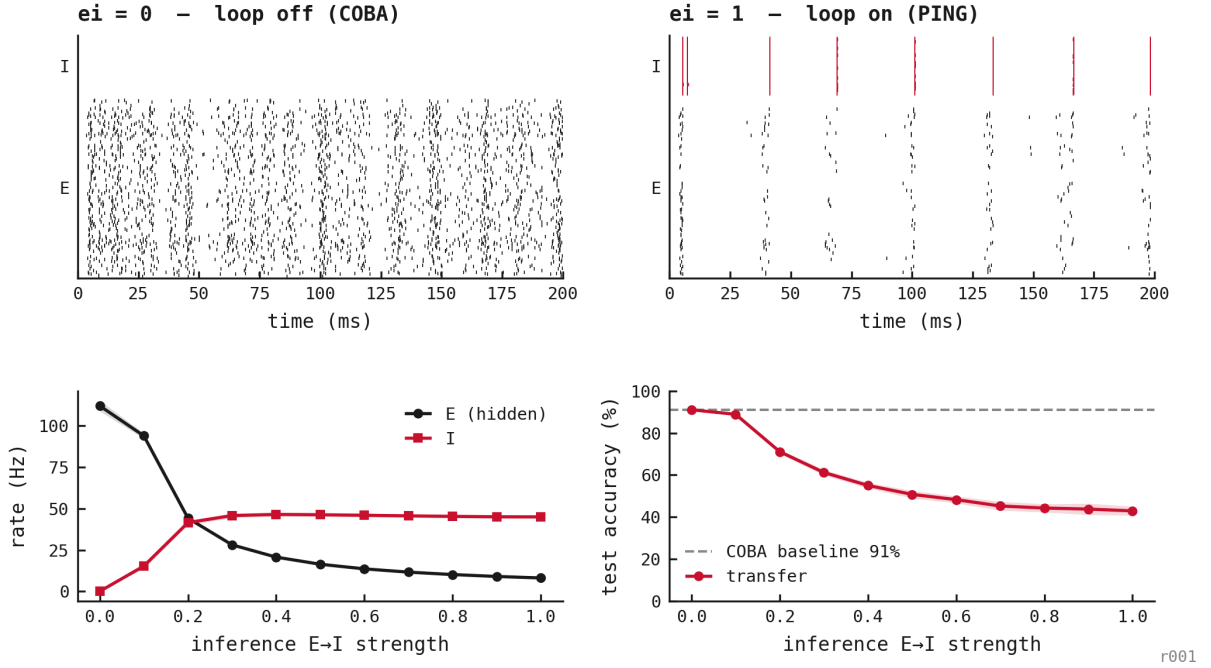
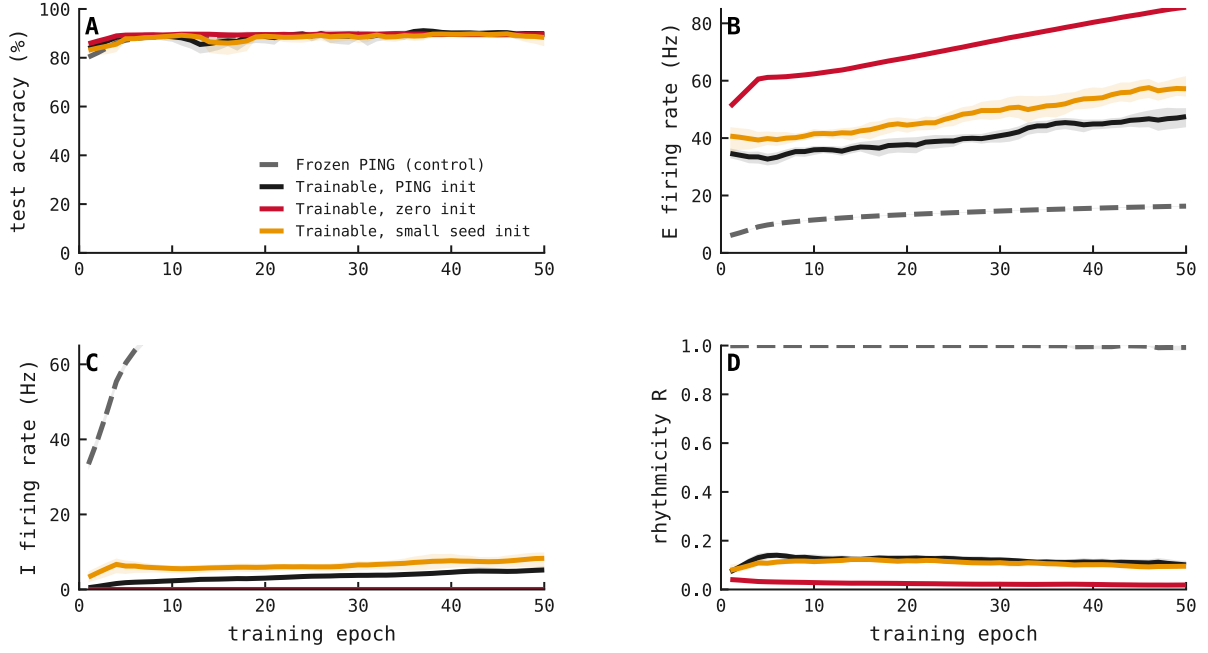
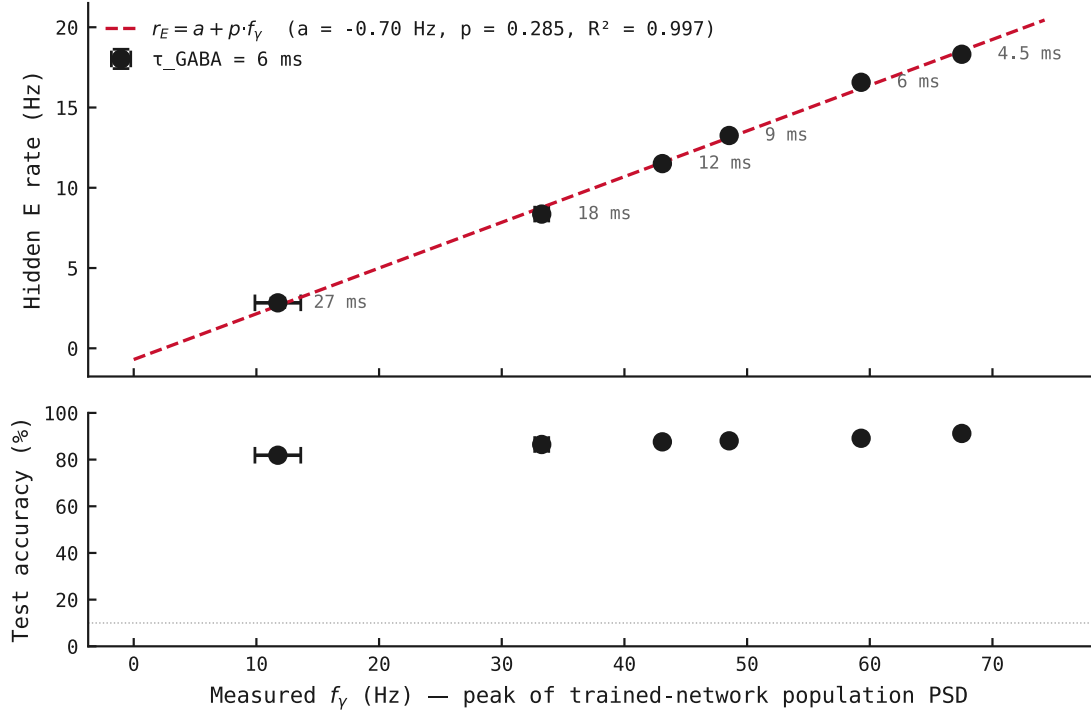


Figure 4: **Engaging the recurrent loop at inference reproduces the PING firing-rate reduction with no weight update.** COBA networks trained with seeds 42–44 are evaluated with recurrent E→I coupling scaled from $ei = 0$ to $ei = 1$, without retraining. Top: illustrative seed-42 rasters at loop-off and full strength. Bottom left: across-seed mean excitatory (black) and inhibitory (red) rates; excitation falls approximately 14-fold (from ≈ 112 to ≈ 8 Hz) as inhibition rises to ≈ 45 Hz. Bottom right: mean accuracy falls from $\approx 91\%$ to $\approx 43\%$ (a ≈ 48 pp cost). Shaded bands show sample SD across seeds. The rate gating appears when the loop is wired in, whereas the accuracy loss reflects the absence of a feedforward compensation learned in the presence of the loop. Source: exp038.



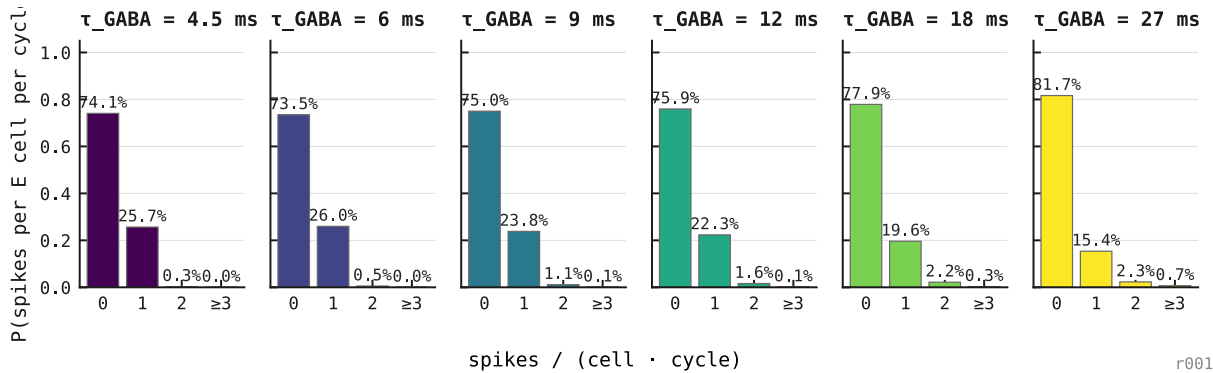
r001

Figure 5: **Releasing the loop weights to gradient descent prunes the rhythm within a single epoch, from every initialisation.** Per-epoch training metrics with the recurrent weights W^{EI} , W^{IE} made trainable under the Dale’s-law clamp, over 50 epochs on MNIST. Lines are the mean of three seeds (42–44); shading is the across-seed range. Conditions differ only in initialisation of the loop weights: canonical PING values (black), zero (red), and $0.1 \times$ canonical (amber), against a frozen-PING control (grey, dashed). **(A)** Test accuracy: all conditions overlap at ≈ 84 – 91% . **(B)** Mean excitatory firing rate: the trainable conditions rise to ≈ 49 – 88 Hz as the loop is dismantled, while the frozen control remains gated near 17 Hz. **(C)** Mean inhibitory firing rate: the trainable conditions collapse to ≈ 0 Hz within a few epochs, while the frozen control’s inhibitory rate rises to ≈ 108 Hz as its readout trains. **(D)** Lobe–trough rhythmicity contrast: the frozen control holds at ≈ 1 while every trainable initialisation drains to ≈ 0.02 – 0.11 . Source: exp049.



r001

Figure 6: **Post-training excitatory rate covaries affinely with gamma frequency across a sweep of inhibitory decay kinetics.** Networks were trained from scratch at each of six values of the GABA decay constant $\tau_{\text{GABA}} \in \{4.5, 6, 9, 12, 18, 27\}$ ms. Changing τ_{GABA} alters both the realised gamma frequency and the duration and integrated influence of inhibition (§5.4). Top: mean post-training excitatory firing rate against measured f_γ ; markers are per-condition means over three seeds, with error bars ($\pm$ SD) on both axes. The linear fit $r_E = -0.7 + 0.285f_\gamma$ passes through every error bar ($R^2 = 0.997$). Bottom: mean test accuracy declines from 91.2% at $\tau_{\text{GABA}} = 4.5$ ms to 81.9% at 27 ms, a 9.3 percentage-point tradeoff. The sweep establishes covariance with realised frequency, not an independent causal effect of frequency alone. Source: exp041.



r001

Figure 7: **The affine rate law follows from a near-binary per-cycle firing statistic: each excitatory cell contributes at most one spike per gamma cycle.** Excitatory spikes were resolved into (cell, cycle) pairs by assigning each spike to the gamma cycle inferred from peaks of the population inhibitory-burst rate (§5.5). Each panel shows the distribution of spikes per pair (0, 1, 2, or ≥ 3) at one value of τ_{GABA} . Across the sweep, $P(0) \approx 75\%$, $P(1) \approx 24\%$, and the multi-spike fraction (≥ 2) stays near $\approx 1\%$; pooled over all conditions, 98.85% of pairs contain at most one spike. Source: exp046.

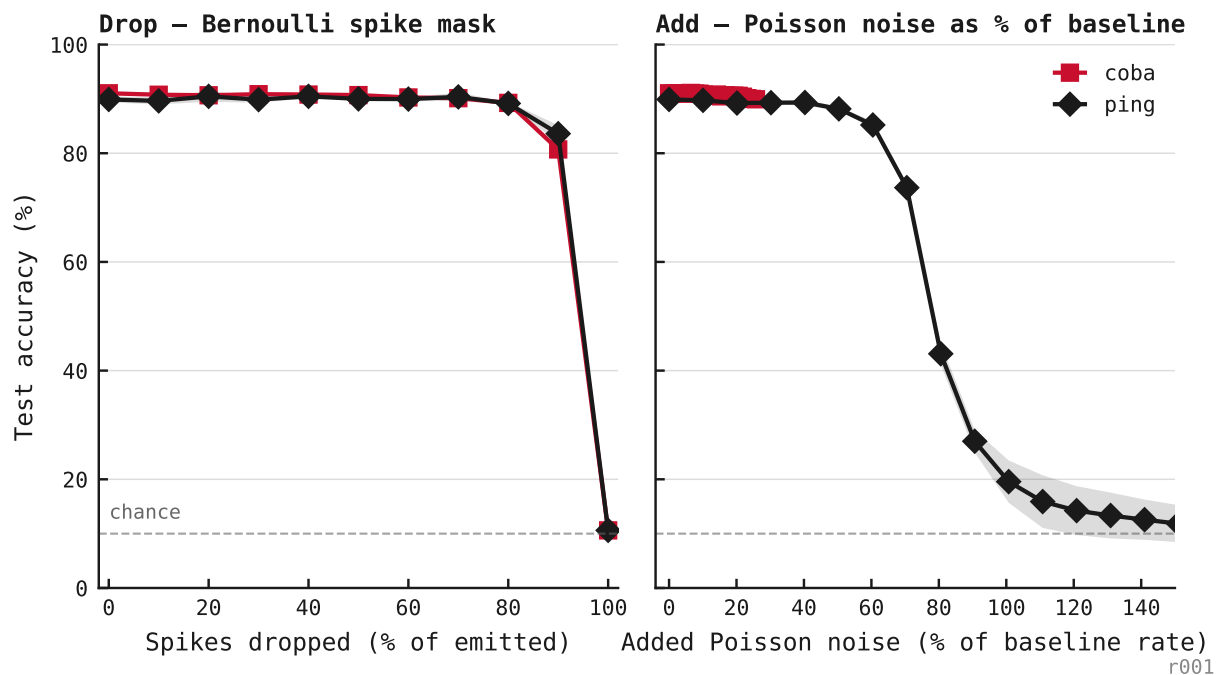


Figure 8: **The gating is robust to spike deletion but fragile to spike addition, the expected signature of a phase-based code.** PING and COBA networks trained with seeds 42–44 are perturbed at inference without retraining. Left: deletion of a random fraction of emitted spikes. PING retains $\approx 89\%$ mean accuracy through 80% deletion, close to its unperturbed value. Right: addition of Poisson spikes, expressed relative to each population’s baseline rate. PING falls toward chance as off-phase spikes disrupt the rhythm; COBA, having no rhythm to protect, is more tolerant to addition. Lines show across-seed means and shaded bands show sample SD. Source: exp037.

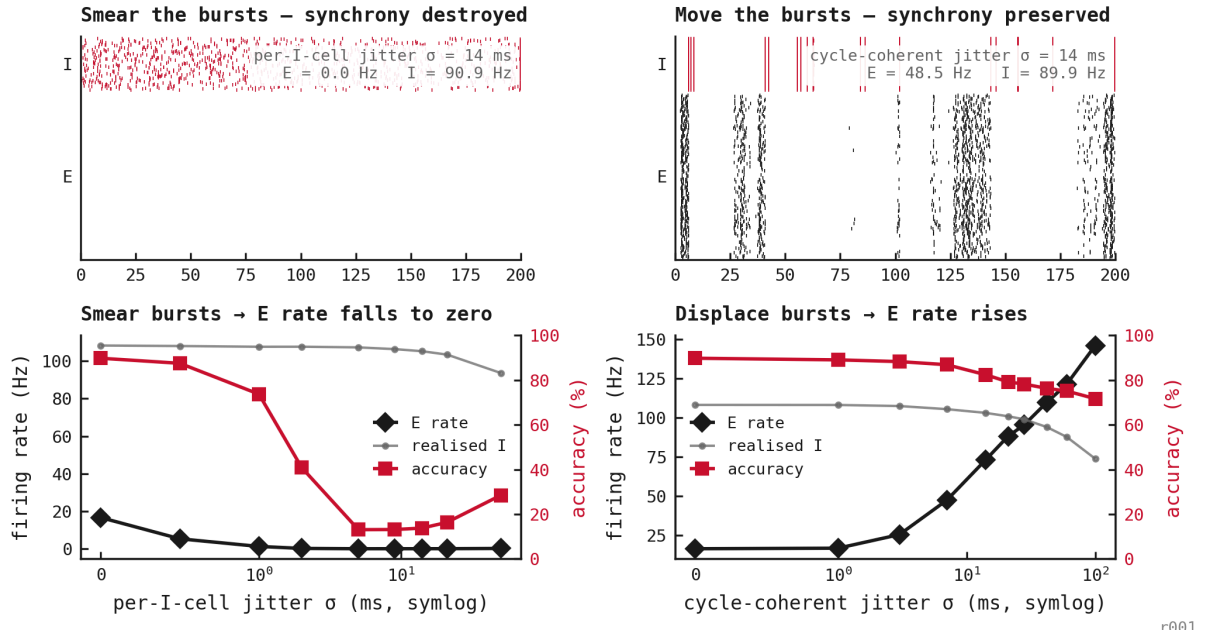


Figure 9: **Two inhibitory-jitter manipulations at the same magnitude that hold the mean inhibitory rate fixed drive the excitatory rate in opposite directions, isolating timing from level.** The trained PING inhibitory spike train is perturbed at inference while the mean per-cell inhibitory rate is held constant. Both arms use the same jitter magnitude, $\sigma = 14$ ms — only the *kind* of jitter differs. Top: single-trial rasters (E black, I red). Bottom: mean excitatory rate (black) and accuracy (red) versus jitter magnitude σ , with the realised mean inhibitory rate overlaid (grey). Left: per-cell jitter smears each burst into a continuous shunt; the excitatory rate falls to ≈ 0 Hz (inhibitory rate ≈ 105 Hz) and accuracy falls to chance. Right: cycle-coherent jitter displaces whole bursts while preserving within-burst synchrony; the excitatory rate rises from ≈ 17 to ≈ 73 Hz (inhibitory rate ≈ 103 Hz, matched to the left arm) and accuracy holds high. Both arms are read where the realised inhibitory rate is matched to within a few percent; at larger σ the cycle-coherent excitatory rate climbs further, but the finite trial window truncates the most-displaced bursts and realised inhibition falls, so the strict comparison is anchored here. Identical mean inhibition, opposite excitatory outcome: the gate is the phase structure of inhibition, not its level. Source: exp042.

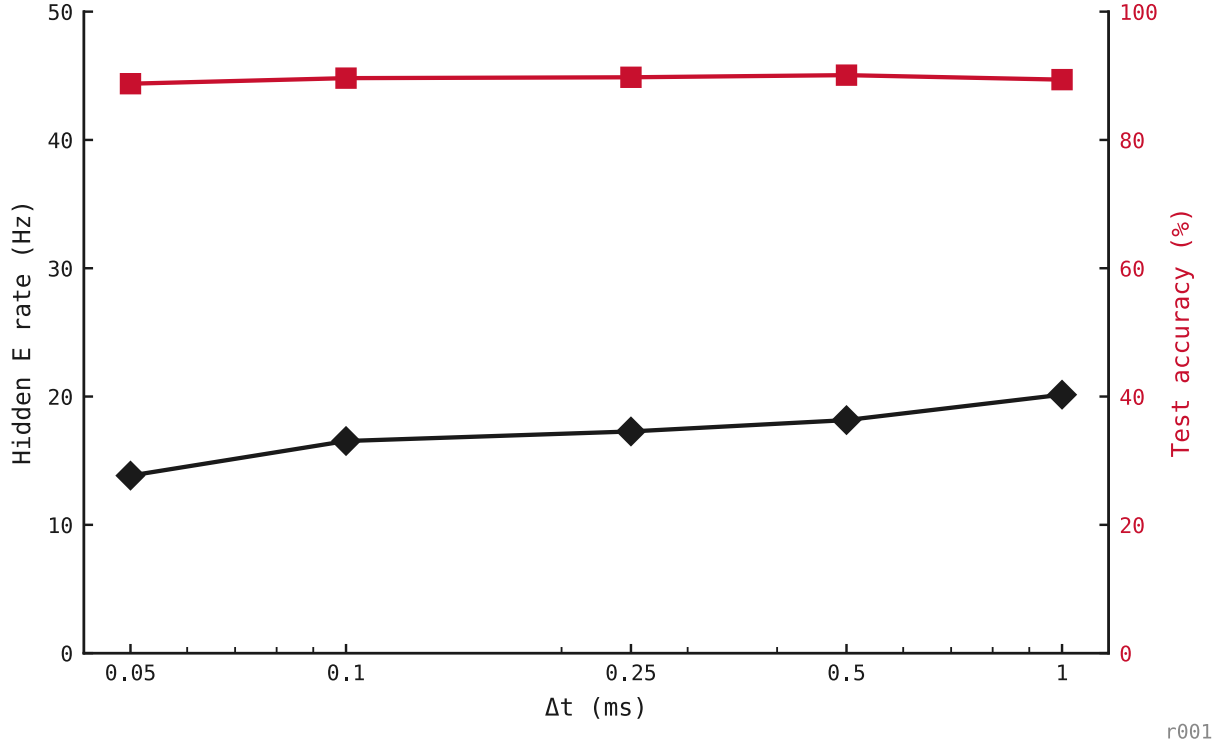
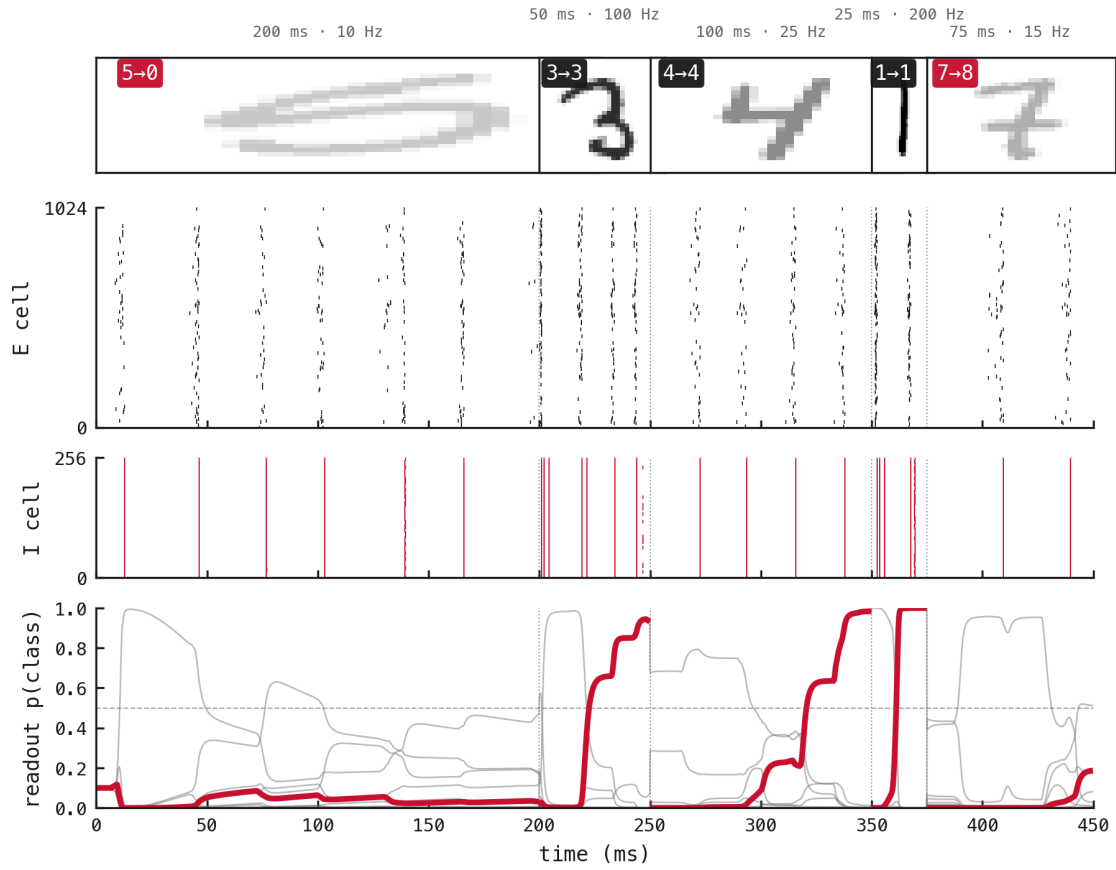


Figure 10: **The firing-rate reduction is a physical-time property, invariant to the integration timestep over a twentyfold range.** The network was trained and evaluated at matched integration timestep Δt for each value across $[0.05, 1.0]$ ms (logarithmic abscissa). Left ordinate (black diamonds): post-training mean excitatory rate, confined to a 13.6–20.2 Hz band and non-monotonic in Δt , so finer stepping does not buy a lower rate. Right ordinate (red squares): test accuracy, flat within 2.2 pp (88.2–90.4%). Because both training and inference use the same Δt at each point, the figure tests invariance of the training-plus-inference pipeline, not the generalisation of one trained network to a varied inference step. Source: exp044.



exp048-replot

Figure 11: **A PING network trained on isolated digits classifies a continuous, unsegmented stream whose presentation timing varies from segment to segment.** A single input stream concatenates five MNIST digits, each with its own presentation duration (25–200 ms) and Poisson input rate (10–200 Hz); the network is given no segmentation cue and is not retrained. Top: the five digit thumbnails with their per-segment (duration, rate) and predicted labels; thumbnail opacity indicates input rate. Below: hidden excitatory and inhibitory rasters, showing gamma cycles maintained throughout (sparser under weak drive, denser under strong drive) and the sliding leaky-integrator readout traces. 3 of 5 digits are classified correctly; the two errors occur in the weakest-drive segments and are interpreted against the population curve in Figure 12B. Source: exp048.

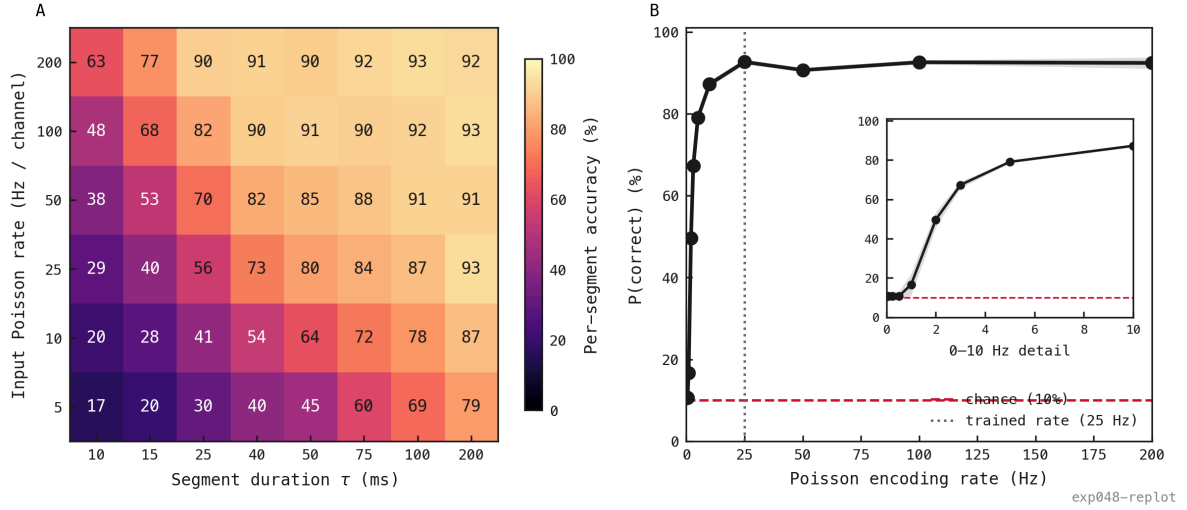


Figure 12: Streaming accuracy has distinct integration-time and encoding-rate evidence floors. (A) Per-segment accuracy across presentation duration τ and input rate, averaged over three seeds and 1,200 segments per grid cell. For $\tau \leq 15$ ms, accuracy does not exceed $\approx 80\%$ at any input rate; above that floor, diagonal iso-accuracy contours show an approximate dependence on $\tau \cdot \text{rate}$. The trained operating point ($\tau = 200$ ms, 25 Hz) reaches 93%. **(B)** Probability of a correct classification versus encoding rate with both presentation duration and readout window fixed at 200 ms. All points belong to the same psychometric curve. Performance remains at chance through 0.5 Hz, is clearly informative by 2 Hz, and reaches 79.1% at 5 Hz; the dotted line marks the 25 Hz training rate. Thus the Figure 11 error at 200 ms and 10 Hz occurs in a condition with 87.3% population accuracy, above the nonviable encoder regime. For scale, $\tau = 15$ ms is approximately 0.7 times the canonical gamma period $T_\gamma \approx 21$ ms, but gamma frequency is not manipulated independently. Source: exp048.